Form 4 Mathematics – ZIMSEC Revision (Paper 1 Style)

Time: 2 hours | Marks: 100

Instructions:

Answer all questions.

Show all working.

Use a calculator where necessary.

Section A: Number and Arithmetic [10 marks]

1. Simplify: $5^2 \times 5^3$ [2]

2. Express 0.000567 in standard form [2]

3. Round 4.6879 to:

a) 1 decimal place [1]

b) 2 significant figures [1]

4. A number is divided among 3 people in the ratio 2:3:5. If the total is 100, find each share. [2]

5. Write 7/12 + 5/8 as a fraction in its simplest form. [2]

Section B: Algebra [10 marks]

6. Solve for x : 3x - 7 = 11 [2]

7. Factorise: $x^2 + 5x + 6$ [2]

8. Simplify: (6x×2y×3)/(3xy) [2]

9. Expand: (x + 3)(x - 2) [2]

10. Solve for y: 2y - 3 = 5y + 6 [2]

Section C: Geometry & Mensuration [10 marks]

11. The length of a rectangle is 12 cm and width 7 cm. Find:

a) Perimeter [1]

b) Area [1]

12. A cylinder has radius 3 cm and height 10 cm. Find the volume ($\pi = 3.142$). [2]

13. Find the exterior angle of a regular hexagon. [2]

14. A triangle has sides 6 cm, 8 cm, and 10 cm. Find its area using Heron's formula. [2]

15. Find the volume of a sphere of radius 5 cm (use $V = 4/3\ \pi\ r^3$, $\pi = 3.142$). [2]

Section D: Trigonometry [10 marks]

16. In a right-angled triangle, $\angle A = 30°$ and the hypotenuse is 10 cm. Find the side opposite $\angle A$. [2]

17. Find $\tan 45°$ [1]

18. A ladder leans against a wall making an angle of 60° with the ground. If the ladder is 8 m long, find how high it reaches on the wall. [2]

19. Find the value of $\sin 30° + \cos 60°$ [2]

20. Solve for x if $\sin x = 0.5$ and $0° < x < 180°$ [3]

Section E: Statistics [5 marks]

21. The marks of 10 students are: 5, 7, 8, 9, 6, 7, 8, 10, 9, 6.

a) Mean [1]

b) Median [1]

c) Mode [1]

d) Range [1]

e) If another student scores 12, find the new mean [1]

Section F: Probability [5 marks]

22. A bag contains 3 red balls, 2 blue balls, and 5 green balls. If one ball is drawn:

a) Probability it is red [1]

b) Probability it is not blue [1]

23. Two coins are tossed. Find the probability of:

a) Two heads [1]

b) At least one tail [1]

24. A die is thrown once. Find the probability of getting an even number [1]

Section G: Linear Equations and Graphs [5 marks]

25. The equation of a line is $y = 2x + 3$.

a) Find the y-intercept [1]

b) Find the gradient [1]

26. Find the equation of a line passing through (2,3) with gradient 4 [2]

27. Determine the point of intersection of $y = x + 2$ and $y = -x + 6$ [1]

Section H: Ratio, Proportion and Rates [5 marks]

28. A map is drawn with scale 1:50 000. If two towns are 8 cm apart on the map, find the actual distance [2]

29. If 5 kg of sugar costs $15, find the cost of 8 kg [2]

30. A car travels 120 km in 2 hours. Find the speed in km/h [1]

Section I: Mensuration of Plane Figures [5 marks]

31. A triangle has base 10 cm and height 8 cm. Find its area [1]

32. A trapezium has parallel sides 6 cm and 10 cm, and height 4 cm. Find its area [2]

33. A square has perimeter 20 cm. Find its area [2]

Section J: Simple Interest & Finance [5 marks]

34. Find the simple interest on $5000 at 8% per annum for 3 years [2]

35. A sum of $2000 is invested at 5% per annum simple interest. Find the total amount after 4 years [2]

36. If a shirt costs $120 and is sold at $150, find the profit percentage [1]

Section K: Indices and Logarithms [5 marks]

37. Simplify $(2^3)^2$ [1]

38. Express 8×10^5 in standard form [1]

39. Solve for x: 2^x = 16 [1]

40. Simplify $9^{1/2}$ [1]

41. Evaluate log 100 (base 10) [1]

Section L: Miscellaneous / Word Problems [10 marks]

42. A tank holds 200 liters of water. 25% of it is used. How much water remains? [2]

43. The sum of two numbers is 45. One number is 5 more than the other. Find the numbers [2]

44. A bus travels 300 km in 5 hours. Another bus travels 420 km in 6 hours. Which bus is faster? [2]

45. The perimeter of a rectangle is 60 cm. If the length is 20 cm, find the width [2]

46. A bag contains 5 red, 3 blue, and 2 green balls. Two balls are drawn without replacement. Find the probability both are red [2]